

# Tensor Omics: Full Toolkit Overview (May 7, 2025)

## A. Data Preprocessing and Representation

- **Normalization:** standard deviation scaling, optional quantile normalization, log transform ( $\log(1+x)$ ); fold-change computation for stress experiments.
- **Expression Vector Construction:** replicate averaging; vectors stored in raw-space and normalized-space K-D Trees.
- **Grid Construction and Smoothing:** automatic grid sizing; interpolation and smoothing of vector fields performed post-space construction and pre-shift calculation.

## B. Semantic Annotation and Axes

- **Metadata:** gene families, duplication metadata, molecular functions, phenotype labels.
- **Anchor Axis:** user-defined semantic reference axis (e.g., time, tissue).
- **Parzival Axes:** lexical contrast vectors (e.g., stress, phenotype modules).

## C. Shift Field and Geometric Observation

- **Shift Vectors:** defined as  $\vec{s} = \vec{p} - \vec{o}$ ; interpolated and smoothed post-estimation.
- **Outlier Detection:** via RDI (distance), RAI (angle), and RAS (axis-based signal).
- **Tissue Specificity:** angle to space diagonal as cosine similarity; versatility analysis.
- **RAP Projection:** Relative Axes Plane for visualizing rotation and preference; subspace RAP for phenotype-limited axes.

## D. Duplication Outcome Analysis

- **Subfunctionalization:** detected via low RDI and RAI of in-paralog sum vector.
- **Neofunctionalization:** detected by RAS — newly activated axes relative to centroid.
- **Dosage Effects:** aligned but amplified in-paralog sum vector (low RAI, high RDI).

## E. Subfield Geometry and Tensor Summary

- **Directional Subfield Selection:** based on metadata, DOI cosine threshold, and background exclusion.
- **Field Descriptors:**
  - Flow vector (first eigenvector of  $C = MM^T$ )
  - Rotation vector (antisymmetric component  $C - C^T$ )
  - Frobenius norm (total signal strength)
  - Rank (via SVD)

## F. Mjöltnir’s Handle — Canonical Axis Reconstruction

- **Purpose:**
  - Defines a smooth, empirically derived semantic axis for subfield tracing and slicing.
  - Serves as the canonical tracing rail for all downstream modules requiring directional coherence.
- **Step 1 — Extract Brokk Axes:**
  - For expression fields: apply PCA to normalized vectors to get spread axes.
  - For shift fields: compute covariance matrix of shift vectors; first eigenvector is dominant flow.
- **Step 2 — Compute Clock Hand Projections:**

$$\vec{v}_i = \vec{r}_i + \vec{v}_i^{\text{clock}}, \quad \vec{r}_i = \langle \vec{v}_i, \vec{b} \rangle \cdot \vec{b}, \quad \vec{v}_i^{\text{clock}} = \vec{v}_i - \vec{r}_i$$

- **Step 3 — Smooth Root Vector Cloud:**

$$\sigma = 1.2 \cdot d_{\text{grid step}}, \quad \text{truncated at } 3\sigma$$

- **Step 4 — Surface Extraction via Eitri 2.0:**
  - Divide Mjöltnir’s handle into equal-length or quantile-based segments.
  - Construct orthogonal planes and form cylinders; select surface candidates by angular binning.

## G. Gradient Visualization: Gwion (formerly GAWD)

*All directional slicing and signal/rank estimation in this module is performed along **Mjöltnir’s handle** (see Section F).*

- **Directional Slicing:** trace field along the dominant flow vector.
- **Panel 1:** Line plot of standardized Frobenius norm and rank along axis.
- **Panel 2:** Clock plot of flow vectors orthogonal to the axis.
- **Panel 3:** Clock plot of rotation vectors, same projection.

## H. Temporal and Trajectory Analysis

*All local plane projections and shift-based trace estimation are guided by **Mjöltnir’s handle**.*

- **Trajectory Modes:** mean-plane projection (Mode A) vs. locally stacked planes (Mode B).
- **Derived Metrics:** Angular Shift Velocity (ASV), Instantaneous Acceleration (IA), vector distance to centroid.

## I. Siddhartha Projection and Surface Tracing

*Multi-dimensional trace visualization and curvature estimation are performed along **Mjöltnir’s handle**.*

- **Multidimensional Trace:** slices a high-dimensional subfield into 3D using its spread axis.
- **Afterglow:** trailing frames visualized with alpha decay and color gradient.
- **Surface Derivatives:** vectors from grid to surface points used to estimate curvature.

## J. Surface Extraction — Eitri

*All surface cylinders are constructed around segments of **Mjölnir’s handle**.*

- **Cylindrical Shells:** projected around Brokk axis; grid-aligned.
- **Clock-hand Analysis:** select max-distance vectors per angular bin within tube slices.
- **Multi-Rail Correction:** check across eigenvector axes; update surface candidates if stronger per sector.

## H. Tensor Field Analysis — Bobito

- **Bobito (Flechita) — Level One:**
  - Cone-based expansion using vector alignment (JOE/JOV)
  - Cutoff detection via derivative jump
- **Bobito (Nubecita) — Level Two:**
  - Subfield grouping via directional coherence of flow/rotation vectors
  - Empirical jump threshold based on sorted cosine similarity

## I. Semantic Glyph Learning — Gilgamesh

- **Gilgamesh Uruk (The Divider):**
  - Trains local SVMs on condition-labeled subfields
  - Tests empirical significance via permutation or bootstrap
  - Enables glyph recognition from semantic separation
- **Gilgamesh Humbaba (The Constellar):**
  - Aligns subfields by Brokk axes
  - Bins or groups by descriptor similarity or phenotype modules
  - Maps higher-order structure from glyph constellations

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## L. Infrastructure and Instrumentation — F42

- **F42 Data Model:** flat-typed arrays, no nested allocatables, portable ‘.txdata’ binary format.
- **Patch System:** timestamped changes with semantic messages; Git- and DVC-compatible.
- **Energy Awareness:** Fortran for numeric kernels, no model training, low-carbon compute philosophy.

## M. Visualization and Documentation — Flyer and Labbook

- **TOX Flyer:**

- Interactive D3-based visualization
- Movie mode: state recording and playback
- JavaScript-based scripting and exportable snippets

- **Labbook:**

- Rendered as HTML
- Embeds Flyer visualizations or animations
- Optional slide deck export (HTML per slide)